Discrete-time systems described by difference equations (FIR and IIR)

Difference equation:

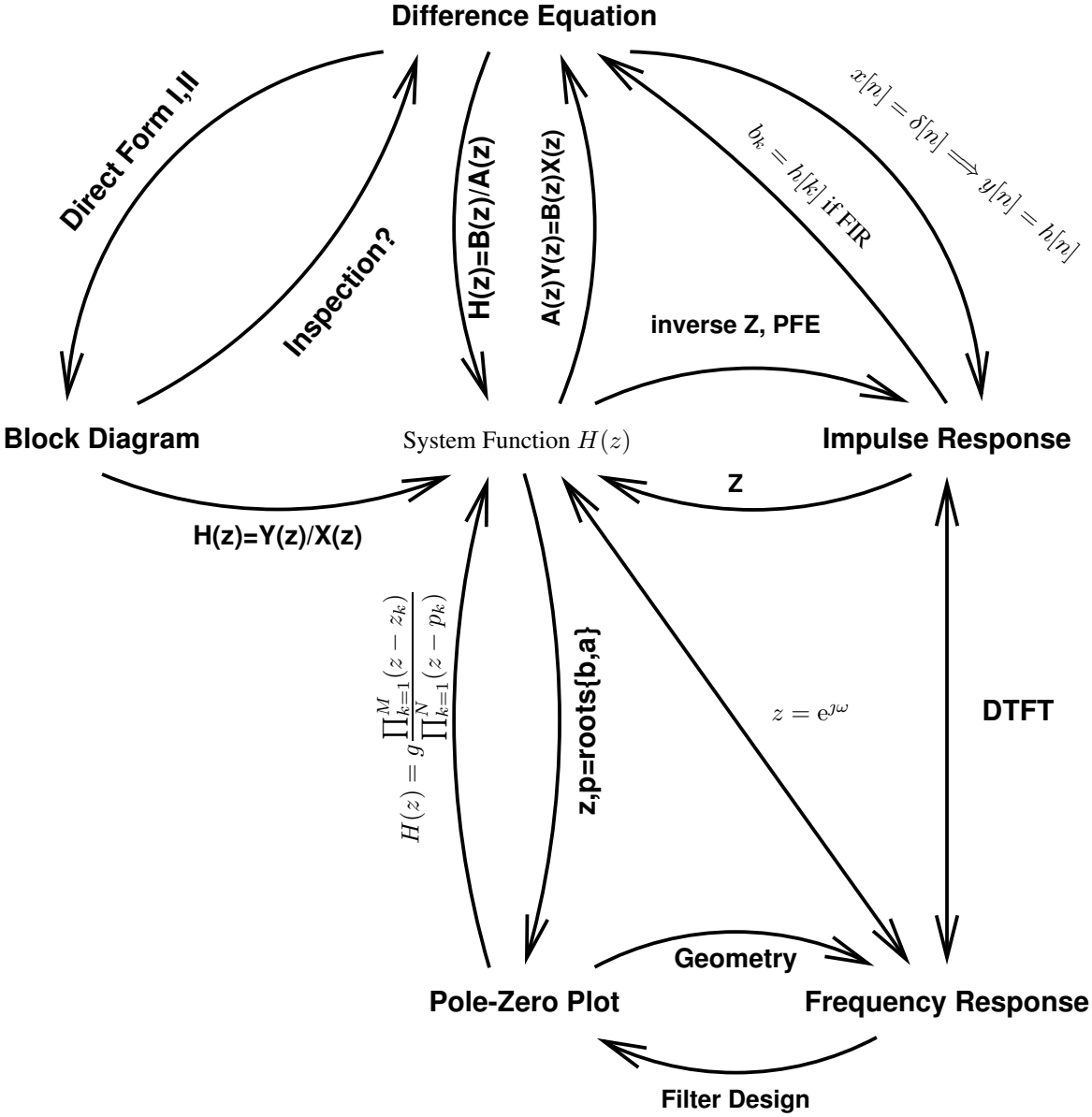
$$y[n] = -\sum_{k=1}^{N} a_k \, y[n-k] + \sum_{k=0}^{M} b_k \, x[n-k]$$

System function (in expanded polynomial and in factored polynomial forms):

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=1}^{N} a_k z^{-k}} = b_0 z^{N-M} \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

Frequency magnitude response: $|\mathcal{H}(\omega)| = b_0 \dfrac{\prod_k |\mathrm{e}^{\jmath\omega} - z_k|}{\prod_k |\mathrm{e}^{\jmath\omega} - p_k|}$

Relationships:



Each representation corresponds to a type of input/output relationship, *e.g.*, convolution.